

CS 4782/5782 Final Report

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Github Repo: <https://github.com/kevinbao86/CS5782-Final-Project-TreeOfThoughts>

Introduction

Paper: Tree of Thoughts: Deliberate Problem Solving with Large Language Models by Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan.

This paper addresses a key limitation of LLMs: standard inference generates text token-by-token from left to right, which can fail on tasks requiring exploration, planning, lookahead, or backtracking. To address this, the authors propose **Tree of Thoughts (ToT)**, a framework that generalizes Chain-of-Thought prompting by allowing the model to explore multiple intermediate reasoning paths, or “thoughts,” rather than committing to one linear chain.

In ToT, the model generates candidate thoughts, evaluates their promise, and uses search algorithms such as breadth-first or depth-first search to continue, prune, or backtrack. The main contribution is showing that LLMs can perform more deliberate problem solving without additional training. On tasks such as Game of 24, Creative Writing, and Mini Crosswords, ToT substantially improves performance; for example, GPT-4 with Chain-of-Thought solved only 4% of Game of 24 problems, while ToT achieved 74%.

Chosen Result & Extension

Our goal is to reimplement the problem solving frameworks described in the paper (standard IO, Chain of Thought (CoT), and Tree of Thoughts (ToT)) and run a LLM on the 3 tasks in the paper (Game of 24, Creative Writing, Crosswords) using these frameworks, in order to see if we can match their experiment performance (i.e. % tasks solved). We also decided to introduce a 4th, more complex, task (4x4 Sudoku) to test the limits of these frameworks. In reality, we were only able to reproduce Game of 24 and Creative Writing due to computational and time constraints.

Methodology

Following the original framework, our implementation consists of four components: thought decomposition, thought generation, state evaluation, and search.

We use Qwen3.6-28B, a 28-billion-parameter open source LLM, through prompting only, without task-specific fine-tuning. The model acts as both the thought generator and state evaluator. For datasets, we use the original ToT repository data for Game of 24, Creative Writing and a public dataset of one million unique 4x4 Sudoku puzzles from Black-Phoenix.

We compare three methods: input-output prompting, chain-of-thought prompting, and Tree of Thoughts prompting. The input-output baseline asks for the final answer directly, while chain-of-thought asks the model to reason step by step. In contrast, ToT explores multiple intermediate reasoning paths before selecting a final solution.

For thought decomposition, Game of 24 thoughts are single arithmetic operations that reduce the remaining numbers. Sudoku thoughts are proposed board updates, such as filling one or more empty cells. For thought generation, the model proposes several possible next steps from the current state. For state evaluation, the model judges whether each partial state is promising: in Game of 24, whether the remaining numbers can still reach 24; in Sudoku, whether the partial grid remains valid and solvable. Our Sudoku implementation also adds explicit rule checking to reject states that violate row, column, or subgrid constraints.

The search algorithm that we used depends on task. For Game of 24 and Creative Writing, we use breadth-first search with breadth $b = 5$. At each step, the system expands candidate thoughts, evaluates the resulting states, and keeps the most promising states for further exploration. This allows ToT to compare alternatives and avoid committing too early to a single flawed reasoning path. For the Creative Writing task, ToT uses a two-level search tree: the first depth generates candidate plans, and the second depth expands those plans into complete passages. In contrast, we used depth-first search for 4x4 Sudoku since DFS has innate backtracking, which is more optimal for solving Sudoku tasks. For this new task, we implemented a new DFS algorithm by ourselves.

Results & Analysis

Method	Success % (Ours)	Success % Paper
IO	6.8	7.3
CoT	3.1	4.0
ToT ($b = 5$)	67	74

Table 1: Game of 24

Method	Coherency Score (Ours)	Coherency Score Paper
IO	1.58	6.19
CoT	3.04	6.93
ToT ($b = 5$)	4.04	7.56

Table 2: Creative Writing

Method	Success %
IO	9.2
CoT	12
ToT	85

Table 3: 4x4 Sudoku

Our re-implementation broadly supports the main finding of the Tree of Thoughts paper: searching over multiple intermediate reasoning paths improves performance over direct prompting and Chain-of-Thought.

For Game of 24, our results closely match the paper. IO prompting achieved 6.8% versus 7.3% in the paper, CoT achieved 3.1% versus 4.0%, and ToT with breadth $b=5$ achieved 67% versus 74%. Although our ToT result is slightly lower, the overall trend is the same: IO and CoT perform poorly, while ToT substantially improves success by generating, evaluating, and searching over multiple candidate reasoning paths.

For Creative Writing, our results preserve the same ordering but differ more in absolute value. IO, CoT, and ToT achieved coherency scores of 1.58, 3.04, and 4.04, compared with the paper’s 6.19, 6.93, and 7.56. This discrepancy is discussed below in the Challenges paragraph.

For our new 4x4 Sudoku task, ToT performed strongly: IO achieved 9.2%, CoT achieved 12%, and ToT achieved 85%. This suggests that ToT is especially effective for constraint-satisfaction tasks, where the model benefits from proposing candidate moves, evaluating partial states, and backtracking from poor choices. However, this result should be interpreted cautiously because Sudoku was not in the original paper and may have used a smaller test sample.

Overall, our results support the paper’s main contribution: LLM reasoning can be improved through inference-time search without additional training. The close Game of 24 reproduction suggests our implementation captures the core ToT mechanism, while the Sudoku experiment shows that the framework can generalize to a new structured reasoning task. Discrepancies, especially in Creative Writing, highlight common re-implementation challenges such as prompt sensitivity, evaluator differences, model choice, and computational limits.

Challenges The original paper’s prompts for Creative Writing were not directly compatible with Qwen3.6, which often failed to follow the expected output format and produced responses our parser could not handle. We therefore rewrote the prompts to be more explicit and directive. The problem was that Qwen frequently inserted extra reasoning text, often inside `<think>` tags, before the final answer. Since this disrupted generation and scoring, we added post-processing to remove reasoning blocks, strip formatting artifacts, extract the final passage, and discard extra self-checking text. This made the outputs more consistent and improved comparison across IO, CoT, and ToT baselines. However, post-processing was not always successful, which likely contributed to our lower Creative Writing scores.

Reflections

This project showed that re-implementing Tree of Thoughts requires more than copying the original prompts. Performance depends heavily on prompt wording, output formatting, parsing, evaluation criteria, and search settings. We found that ToT works best when a task has clear intermediate states that can be generated, checked, and compared. This explains why Game of 24 and 4x4 Sudoku were easier to reproduce than Creative Writing, where quality is subjective and harder to score consistently.

The main takeaway is that inference-time search can improve LLM reasoning without fine-tuning. Our Game of 24 results followed the paper’s trend, and our Sudoku extension showed a large improvement from IO and CoT to ToT. However, Creative Writing revealed the fragility of LLM pipelines: Qwen3.6 often added extra reasoning text or formatting artifacts, requiring prompt changes and post-processing before scoring.

The Sudoku extension also showed that ToT can be especially effective when combined with deterministic rule checking. Rather than relying only on model self-evaluation, symbolic validation helped prune invalid states and made the search more reliable.

Future work could reproduce Mini Crosswords, test larger Sudoku variants, compare different open-source models and search strategies, and improve evaluator design for subjective tasks. Overall, ToT is a promising framework, but its success depends on careful task decomposition, robust formatting, and reliable state evaluation.

References

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